

Rahul Sood

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Objective

As an Architect and Senior Python, Cloud and Big Data engineer, my goal is to leverage my expertise to architect, design, develop, test, and deploy highly available, secure, high-performance, enterprise scale end-to-end solutions for the clients.

Professional Summary

- **Sr Solution Architect at Baxter International / RIPL Technologies**
- Also – **Sr Architect** Python/Cloud/AI/ML (Team size 15-20) at [Kritikal Solutions](#)
- Domain expertise – Finance, Banking, Telecom, Retail, Insurance, SCM, CRM
- Implementing Scrum/Agile methodology using JIRA, Confluence, Azure DevOps(ADO)
- 9+ years in Python/PySpark, Django, FastAPI, Flask, Celery, Pandas, Pytorch, sklearn
- 7+ years in AWS and Azure developing Cloud-native & hybrid applications using Python, Kafka, Kinesis, MSK, RDS, Aurora, DynamoDB, Lambda, Snowflake, Databricks
- Building low-latency, high-throughput applications & Data ingestion pipelines using Azure Data Factory, AWS Glue, Apache Spark/PySpark, Snowflake, Databricks
- Ubuntu / Linux, Apache, Nginx, Gunicorn, uWSGI, SSL, SSO, OAuth2, SAML, OIDC
- Experience in end-to-end ML/DL model development/deployment
- 4+ Big Data: Snowflake, Databricks, Hadoop, HDFS, Spark, Airflow, Kafka, Tableau
- Expertise in Designing, Developing and Scaling secure REST and SOAP Web Services supporting XML / JSON formats
- 10+ years in SQL and RDBMS - Oracle 12c/11g/10g/9i, Postgres - DB Design, Modelling, ETL, Sql Loader, Materialized Views/Partitions, Functions, Procedures, Packages, Triggers, Performance Tuning, Collections, DBMS_LOB, UTL_File

Software Skill Set

Cloud Services AWS	Glue, Redshift, EMR, Dynamodb, Kinesis, S3, RDS, MSK
Cloud Services Azure	Azure Data Factory, Synapse, ADLS, Azure SQL DB
Programming Languages	Python, R, PL/SQL, Javascript, UNIX Shell, PHP
Python frameworks	Django, FastAPI, Flask, Celery, Pytorch, Keras
Big Data & Streaming	Spark, Kafka, Kinesis, Snowflake, Airflow, HDFS, HBase
Database	Oracle, SQL Server, Postgres, Teradata, MySQL, NoSQL
Developer Tools	VS Code, PyCharm, ADO, Git, Postman, JMeter, SoapUI
Misc	Postman, Apache JMeter, Control-M, BJS, Crontab

Work Experience

Jul '24 – Present	Baxter/RIPL: Sr Solution Architect	-	0.5 yrs -	Bangalore
Jul '22 – Jul '23	Accenture/U3 Infotech: Python Engineer	-	1.0 yrs -	Singapore
May '20 – May '22	Kritikal Solns: Senior Architect	-	2.1 yrs -	Bangalore
Aug-16 – May-20	Capgemini: Project Manager	-	3.9 yrs -	Bangalore
Jun-12 – Aug-16	British Telecom: Lead Consultant	-	4.3 yrs -	Bangalore
Jun-10 – May-12	Sears India: Technical Lead	-	2.0 yrs -	Pune
Jun-07 – Jan-09	IBM India: Senior System Engineer	-	1.6 yrs -	Bangalore
Apr-05 – Mar-07	Reuters Asia: Software Developer	-	2.0 yrs -	Singapore
Jul-04 – Apr-05	Optimum/Comtel: Technical Consultant	-	0.9 yrs -	Singapore
Oct-02 – Jul-04	Zensar Technologies: Assoc. Consultant	-	1.9 yrs -	Pune

Education

Bachelors:	B. Tech in Electronics & ECE from IIT Kharagpur
Std. XII:	73% (AISSCE - CBSE) from DAV College, Chandigarh
Std. X:	90% (AISSE - CBSE) from Govt Model Sr Sec School, Chandigarh

Project Experience

Pyspark, Snowflake Data Lake and Databricks Lakehouse

Role: Data Lead and Architect

Environment: AWS/Azure, Python, PySpark, Databricks, Snowflake

Description:

- Technology Lead for building a high-performance Data lakehouse supporting raw data ingestion (batch and streaming) & transform pipelines (ELT) using Databricks, Python, AWS DMS, Kafka, Apache Spark (PySpark)
- Designed End-to-End ETL Pipeline using AWS Step functions, AWS Glue and Snowflake at Baxter. Designed and implemented robust batch and streaming pipelines, integrating with diverse data sources such as the S3 data lake, Kafka streams, and Postgres Change Data Feed using AWS DMS. Orchestrated end-to-end production pipelines effectively through Databricks Workflows and Cron scheduling, ensuring timely and efficient data processing. Leveraged Databricks Repos with Github for version control. Implemented Delta Sharing for improved data collaboration and incorporated effective data governance practices. Applied Spark SQL in conjunction with Auto Loader to effortlessly ingest and process thousands of JSON and CSV files per day, achieving high data throughput and performance. Worked with Fixed schema and Flexible Schema. Incorporating Custom business logic using Spark SQL like Merge statements, aggregate functions, user-defined functions. The lakehouse aggregated tables served as a source for building parameterized dashboards. Tracking the Data lineage of ingested data down to individual files and timestamps. Sometimes rolling back or segregating the ingested files selectively.
- Built End-to-End Data Ingestion Pipelines using API Gateway, S3, SQS, Lambda, Kinesis Firehose & Snowflake
- Designed ETL pipelines with Azure Data Factory with Databricks integration.
- Built a low-latency, high throughput stream processing pipeline to ingest Network Logs using S3, Azure databricks lakehouse and Kafka cluster.
- Change Data Capture (CDC) using PostgreSQL, MSK, MSK Connect, Debezium, Kinesis Firehose, S3, Snowflake
- Set up data ingestion pipelines to load CSV, Json and Parquet file formats from AWS S3 data lake using Snowpipe
- Optimized the design by incorporating patterns like Schema evolution, Dead Letter Queues, Idempotent consumers, Auto-retries, Buffers/Throttling, transient EMR clusters.
- Setup Multi-cluster warehouse using Snowflake and implemented Materialized views resulting in upto 90% reduction in query execution time.
- Orchestrated over 30 data workflows with Apache Airflow, reducing manual effort by 75% and ensuring on-time data processing.
- Designed Batch and Incremental Load ETL pipelines using AWS Glue jobs and Bookmark with Spark as the backend, S3 as the source data lake and Redshift as the target Data Warehouse
- Loading Unstructured data into Snowflake using Variant data type and Parse_json
- Apache Airflow for workflow orchestration and monitoring of Spark jobs.
- Setup managed Apache Spark / PySpark clusters on Azure and AWS
- Utilized PySpark to process and analyze large datasets, improving data quality and accuracy by 20%.
- Migrated ETL workloads from nightly batch jobs in On-prem SSIS to Spark cluster on Cloud to process high volume of transactional data.
- Business Analyst for a team of 8 people for building high performance ETL pipelines using Datastage and interactive BI Reporting dashboards using Microstrategy
- Rich working exposure to Tableau and PowerBI

Python & AI/ML experience

Role: ML Engineer and Lead

Environment: AWS/Azure, Python, Keras, TF, PyTorch, NLTK, Flask, sklearn

Description:

- 7+ years of experience in Python; mentored team on Python and ML/DL
- Deployed ML models into Production as REST APIs: Postman/AWS/Flask/Apache
- Worked on ML use-cases on Regression, Classification, and Computer Vision problems like Image classification project using Convolutional Neural Networks (CNN), Deep Learning, Transfer Learning, Dropout, OpenCV, Keras, Tensorflow, PyTorch, Python Pandas, Python Iterators, Generators, Numpy, Scipy, skimage
- In-depth knowledge and understanding of famous Deep Learning Models like VGG16, ResNet50, AlexNet, Inception, RCNN, YOLO
- Co-authored and presented a whitepaper on using Deep Learning to estimate the damages from input images in vehicle insurance industry
- Authored a blog titled "*Recommender Systems using Apriori*"

Business Problems Solved:

Deep Learning:

- Model for Grading of Cars based on external damage for Auto Auctions client(IAAI)
- Image Similarity Search / Visual Search with Transfer Learning for eCommerce client
- Using Azure Custom Vision AI Service for training Computer Vision models
- Face Authentication from Video feed using Histogram of Gradients method (HOG)
- Multi-label Image Classification using Deep Learning (CNN in Keras)
- Template Matching, Edge detection, Corner Detection, Image Convolution using OpenCV, Face detection using Voila Jones Classifiers / Haar Cascades

NLP:

- Built a model for Multi-label Text Classification for Healthcare client using Doc2Vec
- Created a model for predicting User Ratings from Restaurant reviews
- Performed NER on Twitter streams; Tweet clustering
- Text Summarization using sentence similarity and textrank algorithm
- Document Classification(unsupervised) using Topic modelling (LDA) in gensim
- Sentiment Analysis based on TF-IDF scores using Multinomial Naïve Bayes

Python, Machine Learning and Time Series:

- Document Search Engine using Inverted Indexes and Linear Algebra
- Predicting weekly Sales for retail stores using multiple regressors
- Predicting customer churn using ANN and Random Forest
- Recommender Systems using SVD, Apriori algorithm, Python Iterators / Generators
- Sales forecasting using ARIMA Time Series modeling
- Identifying location for setting up New Retail Stores using K-means clustering

Responsibilities:

- Evaluate datasets, identify attributes for predictive models for a business problem
- Working closely with data engineering team; identifying and agreeing on *sources, method, formats, encoding, frequency, volume, quality* of data collection
- Data pre-processing, Image pre-processing, data transformations (feature engineering), EDA, training ML models, evaluating model performance, Hyperparameter tuning / Grid-Search, Code optimization
- Communicating insights to stakeholders with Matplotlib/Seaborn/Plotly visualizations
- Compiling training material, whitepapers/blogs; mentoring and supporting the team

Professional Training / Certifications

- Databricks Data Engineer Associate
- SnowPro – Snowflake Certified Developer - [Badge](#)
- Databricks certified Apache Spark (PySpark) Developer - [Badge](#)
- Astronomer certified Apache Airflow Developer - [Badge](#)
- Oracle Certified Associate (OCA) in SQL / PL/SQL
- Deep Learning with PyTorch on Udacity
- Tableau Training on Udemy 2018 – [Certificate](#)
- Data Science using Python on Simplilearn 2018 – [Certificate](#)
- Machine Learning and Data Science in Python on Udemy 2019 – [Certificate](#)
- Python and Data Science iCompass Certificate at Capgemini
- Angular 4+ Training at Capgemini – Top 10 among 200 Candidates
- Oracle Performance Tuning training at British Telecom
- Cross-Cultural Corporate Training at Zensar

Achievements & Interests

- Attending Snowflake and Databricks conferences actively
- Won First Prize in Whitepaper challenge in Automation Hackathon at Capgemini with paper on Vehicle Damage Assessment using Deep Learning
- Got 2nd Prize in Men's Doubles at Sears Table Tennis Championship
- Got 3rd Prize as Inter-Hall Choreography Captain at IIT
- Participated in 5.6 km JP Morgan Chase Corporate Challenge 2006
- Wrote and directed a play for Open-IIT Hindi Dramatics Competition at IIT.